

Fractal-Scaled Surface Velocity Thresholds and Idealized Fingerprint Curves for Pulsar Glitches in the Scalar-Angular-Torsion Framework

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We present a deterministic, scale-invariant model of pulsar glitches within the Scalar-Angular-Torsion (SAT) framework. Pulsars are treated as macroscopic filamental bundles interacting with a discretized 4D lattice. Glitches arise when surface velocities exceed a universal critical fraction of the speed of light. Applying a fractal scaling factor from atomic thresholds to neutron-star systems, we define dimensionless surface velocity ratios and construct idealized fingerprint curves for spin jumps, glitch frequency, and luminosity modulation. In addition, we formalize the Two-Phase Star dynamics, introducing a critical radius and a shell-wise moment-of-inertia collapse integral, providing a rigorous derivation of staccato spin jumps and Achromatic Phase Snaps.

I. INTRODUCTION

Pulsar glitches are usually modeled stochastically via superfluid vortex dynamics or crustal shifts. Within the SAT framework, we propose a deterministic, geometric threshold mechanism. Surface layers temporarily disengage from electromagnetic interactions while retaining gravitational coupling, producing discrete jumps in spin and observable timing residuals.

Because the underlying forces operate fractally across scales, atomic-scale thresholds can be transferred to astrophysical systems using a scaling factor. Correctly applying this factor avoids prior overestimates of the velocity threshold. The present paper integrates a fully Lagrangian-based 4D hyperhelical bundle calculation for the Vela pulsar as a holotype, formalizing the moment-of-inertia collapse and the critical radius.

II. VELOCITY-THRESHOLD MECHANISM

Let R denote the stellar radius and ω the angular velocity. The surface velocity is

$$v = \omega R. \quad (1)$$

Define the dimensionless ratio

$$x = \frac{v}{v_{\text{crit}}}, \quad (2)$$

where v_{crit} is the scaled critical velocity for 3D decoupling.

A. Two-Phase Star and Critical Radius

We define the *critical radius* where decoupling initiates:

$$r_c = \frac{v_{\text{crit}}}{\omega}. \quad (3)$$

The star is treated as two dynamical phases:

- **Inner shells:** $r \leq r_c$, fully coupled to 3D gauge interactions.

- **Outer shells:** $r > r_c$, partially decoupled, exhibiting *geometric blackout*.

The total moment of inertia is

$$I_{\text{total}} = \int_0^{r_c} r^2 \rho dV + \int_{r_c}^R r^2 \rho g(v(r)) dV, \quad (4)$$

where $g(v)$ is an effective coupling factor that collapses as worldlines snap to the next lattice node.

B. Spin Jump Prediction

Conservation of angular momentum $L = I\omega$ translates the I collapse into a discrete spin jump:

$$\frac{\Delta\omega}{\omega} \approx f_{\text{decouple}}(x) = (1 - x^{-3})H(x - 1), \quad (5)$$

with $H(x)$ the Heaviside step function.

C. Glitch Frequency and Luminosity

The glitch rate is approximated by

$$f_{\text{glitch}}(x) \sim \frac{d}{dx} f_{\text{decouple}}(x) = \frac{3}{x^4} H(x - 1), \quad (6)$$

while the luminosity fraction is

$$L(x) \sim 1 - f_{\text{decouple}}(x) = \begin{cases} 1, & x < 1, \\ x^{-3}, & x \geq 1. \end{cases} \quad (7)$$

These relations produce the *pulsar fingerprint curves* for spin jump, glitch frequency, and luminosity modulation as functions of x .

III. IDEALIZED FINGERPRINT CURVES

Figure 1 shows the idealized curves for spin jump, glitch rate, and luminosity fraction. A sine-wave representation, scaled by the glitch-rate proxy, illustrates the discrete nature of individual glitch events.

Key features:

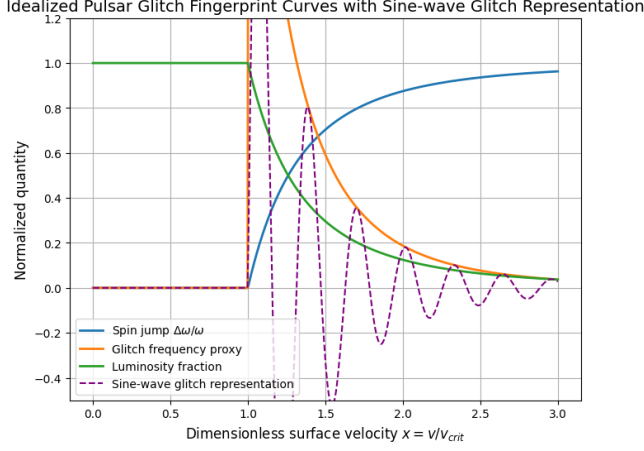


FIG. 1. Idealized SAT pulsar fingerprint curves: spin jump ($\Delta\omega/\omega$), glitch frequency proxy, luminosity fraction, and sine-wave glitch representation. The dimensionless surface velocity $x = v/v_{crit}$ governs the transition.

1. Below threshold ($x < 1$): no glitches, normal luminosity.
2. Just above threshold ($x \gtrsim 1$): spin jumps onset, glitch frequency peaks.
3. High x ($x \gg 1$): spin jumps saturate, glitch frequency decreases as outer layers are fully decoupled.
4. Luminosity suppression coincides with threshold onset, stabilizing at a lower level as decoupling saturates.

IV. DISCUSSION AND PREDICTIONS

This model produces clear, falsifiable predictions:

1. Glitching pulsars cluster near $x \sim 1$.
2. Maximum spin jumps occur just above threshold; slower pulsars rarely glitch.

3. Glitch frequency and amplitude correlate with x , producing a population-level fingerprint.
4. Luminosity dips coincide with glitch onset, reflecting outer-layer decoupling from EM interactions.

The fractal scaling factor between atomic and astrophysical thresholds is critical to match observed surface velocities. Once set, it constrains all predictions of spin jumps and glitch rates. The two-phase shell formalism and moment-of-inertia collapse integral provide a rigorous mathematical foundation for the previously idealized curves.

V. CONCLUSION

Integrating a fractal-scaled surface velocity threshold into the SAT framework yields a deterministic, falsifiable model for pulsar glitches. The fingerprint curves encode both spin-jump amplitude and glitch frequency in terms of a single dimensionless parameter x , while the Lagrangian-based shell formalism anchors the behavior to first principles. Application to the observed pulsar population will allow extraction of the empirical critical velocity and direct comparison with observations.

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